

Polynomial Chaos Concentration: the General Theorem Behind Nelson's Estimate

Polynomial Chaos Concentration: the General Theorem Behind Nelson's Estimate

Drafted 2026-05-08. The framing as a large-deviations / polynomial-chaos concentration problem was suggested by Arthur Jaffe in conversation in his office (and is awaiting his vetting of this writeup). Two-pass vetted by Gemini deep-think (2026-05-08): pass 1 caught a fatal integration bug in the original §4 (the polynomial-chaos bound is symmetric in $|F|$, but V_a is asymmetric, so the direct integration $\int \exp(2|V_a|)$ diverges); pass 2 approved the corrected version below, which retains the full Glimm–Jaffe / Simon dynamical-cutoff structure and applies polynomial-chaos concentration only to the rough error E_R .

Context

Nelson's exponential estimate $\int e^{-2V_a} d\mu_{\text{GFF}} \leq K$ (uniform in lattice spacing a ; this is the cutoff used by Glimm–Jaffe Ch. 8 / Simon Ch. I) is the last analytic input still axiomatized in pph12 after the Phase 2 / PR #14 discharges. Arthur Jaffe (in conversation in his office) confirmed the proof method is large deviations: the dynamical-cutoff trick (split $C = C_S(T) + C_R(T)$, choose $T = T(M)$ to balance the smooth lower bound against the rough-error tail) is a saddle-point / Legendre-transform optimization in the LD sense, and the underlying tail bound is a polynomial-chaos concentration inequality.

If we formalize the abstract concentration theorem, all four Cluster A axioms collapse to short corollaries.

The abstract theorem

Let $(\Omega, \mathcal{F}, \mathbb{P})$ carry a centered Gaussian process $(g_x)_{x \in X}$ (X finite or infinite). For $d \in \mathbb{N}$, let $\mathcal{H}^{\leq d}$ denote the closed subspace of $L^2(\mathbb{P})$ spanned by polynomials of total degree $\leq d$ in the g_x . (Wiener chaos: $\mathcal{H}^{\leq d} = \bigoplus_{k \leq d} \mathcal{H}_k$, where $\mathcal{H}_k$ is the k -th homogeneous chaos.)

Theorem (polynomial chaos concentration). *There is a universal constant $c_d > 0$ depending only on d such that for every centered $F \in \mathcal{H}^{\leq d}$ (i.e. $\mathbb{E}F = 0$):*

$$\mathbb{P}(|F| > \lambda \|F\|_{L^2}) \leq 2 \exp(-c_d \lambda^{2/d}), \quad \lambda > 0.$$

Equivalently:

$$\|F\|_{L^p} \leq (p-1)^{d/2} \|F\|_{L^2}, \quad p \geq 2 \quad (\text{Bonami–Nelson}).$$

The two forms are equivalent up to multiplicative constants by Markov + optimization.

The exponent $2/d$ is sharp: take $F = g_1^d$ with $g_1 \sim \mathcal{N}(0, 1)$. Then $\|F\|_{L^2} = \sqrt{\mathbb{E}[g_1^{2d}]}$ and $\mathbb{P}(|g_1^d| > \lambda \|F\|_{L^2}) = \mathbb{P}(|g_1| > (\lambda \|F\|_{L^2})^{1/d})$ matches the stated decay.

Proof sketch (three ingredients)

Ingredient 1: hypercontractivity of OU

For the Ornstein–Uhlenbeck semigroup T_t on $(\Omega, \mathbb{P})$ (with stationary distribution the Gaussian process):

$$\|T_t f\|_{L^p} \leq \|f\|_{L^q}, \quad e^{2t} \geq \frac{p-1}{q-1}, \quad 1 < q \leq p < \infty.$$

Restricted to homogeneous chaos $\mathcal{H}_k$, the semigroup T_t acts by e^{-kt} (eigenfunction expansion), so for $f \in \mathcal{H}_k$:

$$e^{-kt} \|f\|_{L^p} \leq \|f\|_{L^q}, \quad \text{i.e.} \quad \|f\|_{L^p} \leq e^{kt} \|f\|_{L^q}.$$

Choosing $q = 2$ and $e^{2t} = p - 1$:

$$\|f\|_{L^p} \leq (p-1)^{k/2} \|f\|_{L^2}, \quad f \in \mathcal{H}_k, \quad p \geq 2.$$

Triangle inequality across $k = 0, \dots, d$ extends this to $\mathcal{H}^{\leq d}$ with constant $(p-1)^{d/2}$ (up to constants depending on d).

Ingredient 2: Markov / Chebyshev

For any $\lambda, p > 0$:

$$\mathbb{P}(|F| > \lambda \|F\|_{L^2}) \leq \frac{\mathbb{E}|F|^p}{(\lambda \|F\|_{L^2})^p} = \left(\frac{\|F\|_{L^p}}{\lambda \|F\|_{L^2}} \right)^p.$$

Ingredient 3: optimize p

Combining 1 + 2:

$$\mathbb{P}(|F| > \lambda \|F\|_{L^2}) \leq \left(\frac{(p-1)^{d/2}}{\lambda} \right)^p = \exp\left(p \left[(d/2) \log(p-1) - \log \lambda \right]\right).$$

Set $p-1 = (\lambda/e)^{2/d}$ (so $(d/2) \log(p-1) - \log \lambda = -d/2$):

$$\mathbb{P}(|F| > \lambda \|F\|_{L^2}) \leq \exp\left(-(d/2) (\lambda/e)^{2/d}\right), \quad \lambda \geq e^{d/2}.$$

For $\lambda < e^{d/2}$ the bound is trivial (≤ 2). Tightening the constant gives the cleaner form $\mathbb{P}(|F| > \lambda \|F\|_{L^2}) \leq 2 \exp(-c_d \lambda^{2/d})$.

Specialization to Cluster A

Retraction (2026-05-08, after Gemini vetting). An earlier draft of this section claimed Cluster A could be discharged by integrating the tail of the polynomial chaos concentration theorem applied to all of V_a — i.e. $\int \exp(2|V_a|) \leq 2 + \int_0^\infty 2e^{2\lambda} \cdot 2 \exp(-c(\lambda/\|V_a\|_2)^{1/k}) d\lambda$. This integral diverges: for $k = 2$ (the standard ϕ^4 case) the integrand is $\exp(2\lambda - c\sqrt{\lambda})$, in which the linear 2λ term dominates the stretched-exponential decay. The issue is that the polynomial-chaos bound is symmetric in $|F|$, but V_a is highly asymmetric: it is bounded below modulo Wick shifts (so its left tail is strongly suppressed), while the right tail of V_a is genuinely heavy. A symmetric bound is far too weak to control $\int \exp(-2V_a)$, where what matters is the left tail of V_a alone.

The conclusion: polynomial-chaos concentration is one of three ingredients in the Glimm–Jaffe Ch. 8 / Simon Ch. I dynamical-cutoff proof, not a shortcut around it. The four Cluster A axioms still need the full small-field / large-field decomposition.

The corrected route to discharging Cluster A:

1. Decompose the field: $\phi = \phi_S + \phi_R$ via the covariance-split $C = C_S(T) + C_R(T)$ already proved in `CovarianceSplit.lean`.
2. Smooth lower bound: $V_S(\phi_S) \geq -C_1(1 + |\log T|)^2$ (small-field classical bound, no probability — already proved as `smooth_interaction_lower_bound_log` in `SmoothLowerBound.lean`). The smoothness of ϕ_S makes the Wick polynomial a *bounded-below* classical function on ϕ_S .
3. Rough error tail bound: $E_R = V(\phi) - V_S(\phi_S)$ is a degree- $2k$ Wick polynomial in the *rough* field ϕ_R . Apply polynomial chaos concentration to E_R only:

$$\mathbb{P}(|E_R| > \lambda) \leq 2 \exp(-c \lambda^{2/(2k)} T^{-\delta/(2k)}),$$

the $T^{-\delta/(2k)}$ factor coming from $\|E_R\|_2 \sim T^{\delta/2}$ (the rough-covariance estimate already in `CovarianceSplit.lean`).

4. Dynamical cutoff (the saddle-point / large-deviations step): given a target deviation level M , choose $T = T(M)$ such that $C_1(\log T)^2 = M/2$, i.e. $T = \exp(-\sqrt{M/(2C_1)})$. Then $V(\phi) \leq -M$ forces $|E_R| \geq M/2$, so

$$\mathbb{P}(V \leq -M) \leq \mathbb{P}(|E_R| \geq M/2) \leq \exp(-c' \exp(c'' M^{1/2})),$$

doubly exponential. Integrating $\int_0^\infty \exp(2M) \cdot \mathbb{P}(V \leq -M) dM < \infty$ gives the uniform bound on $\int \exp(-2V_a) d\mu_{\text{GFF}}$.

Mapping to the existing scaffolding. The four pphi2 Cluster A axioms are recovered by the four flavours of μ_{GFF} we use (symmetric lattice, asymmetric lattice, and their hypercontractive-moment forms). The polynomial-chaos concentration theorem above replaces the `True-stub` bodies of `rough_error_Lp_bound` / `rough_error_tail_bound` in `RoughErrorBound.lean`; the small-field side is already proved; the $T = T(M)$ optimization and the Schwinger-style integral $\int \exp(2M) \mathbb{P}(V \leq -M) dM$ are the new content of `NelsonEstimate.lean`.

In the existing scaffolding, the sub-conclusions are:

pphi2 file	scaffolded statement	role in the LD argument
<code>RoughErrorBound.lean</code>	$\ E_R\ _{L^p} \leq C p^d T^{\delta/2}$	hypercontractivity (Ingredient 1)
<code>rough_error_Lp_bound</code>		on the rough chaos component
<code>RoughErrorBound.lean</code>	$\mathbb{P}(E_R > \lambda) \leq$	Markov + optimize (Ingredients 2
<code>rough_error_tail_bound</code>	$\exp(-c \lambda^{2/d} T^{-\delta/d})$	+ 3)
<code>SmoothLowerBound.lean</code>	$V_S \geq -C_1(1 + \log T)^2$	classical small-field bound (no LD)
<code>smooth_interaction_lower_bound_log</code>		
<code>NelsonEstimate.lean</code> (currently axiom)	$\int \exp(-2V) d\mu_{\text{GFF}} \leq K$	dynamical cutoff: tune $T = T(M)$, integrate

After the LD theorem lands, all four `True-stub` theorems in `RoughErrorBound.lean` become specializations, and the `NelsonEstimate.lean` assembly is just an integral against the tail bound.

Notes on infinite-dimensional generalisation

For pphi2 the lattice has finitely many sites, so we never leave the finite-dimensional Gaussian setting. The infinite-dimensional version (X separable Hilbert space, (g_x) a centered Gaussian on it) holds verbatim — Jan-son’s *Gaussian Hilbert Spaces* (1997) Ch. 5 gives the abstract framework. The c_d constants are independent of $\dim X$ because the proof is via OU hypercontractivity, which holds in arbitrary dimension.

The relevance for the continuum limit: when we eventually do $a \rightarrow 0$, V_a lives in a fixed-rank chaos (degree $2k$ is fixed by the polynomial P), but the underlying Hilbert-space dimension explodes. The constant $c_d = c_{2k}$ is dimension-independent, which is why the uniform bound survives.

Connection to Mathlib

- `Mathlib.Analysis.MeanInequalities` — Hölder/Markov machinery.
- `Mathlib.MeasureTheory.Function.LpSpace` — L^p infrastructure.
- `Mathlib.Probability.Distributions.Gaussian` — Gaussian random variables.
- `Mathlib.Analysis.Hypercontractivity` — does not yet exist. Could be contributed alongside this work.
- The markov-semigroups sister project has Bakry-Émery / Brascamp-Lieb infrastructure for the OU semigroup that overlaps. Worth a coordination pass before starting.

References

- A. Bonami, *Étude des coefficients de Fourier des fonctions de $L^p(G)$* , Ann. Inst. Fourier 20 (1970). Original $(p-1)^{1/2}$ hypercontractivity for one Gaussian variable.
- E. Nelson, “The free Markov field,” *J. Funct. Anal.* 12 (1973). Generalises to OU semigroup; the construction step uses exactly the bound in question.
- J. Glimm and A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, Springer (1987), Ch. 8 (dynamical cutoff for $P(\varphi)_2$).
- B. Simon, “The $P(\varphi)_2$ Euclidean (Quantum) Field Theory”, Princeton (1974), Ch. I (cleanest exposition of the dynamical-cutoff trick).
- S. Janson, *Gaussian Hilbert Spaces*, Cambridge (1997), Theorem 5.10 and its corollaries. The abstract concentration theorem in modern form.
- R. Adamczak and P. Wolff, “Concentration inequalities for non-Lipschitz functions with bounded derivatives of higher order,” *Probab. Theory Related Fields* 162 (2015), 531–586. Modern polynomial-chaos concentration with explicit constants.
- B. Maurey, *Some deviation inequalities*, Geom. Funct. Anal. 1 (1991). An LDP-style perspective on polynomial chaos via entropy arguments.

Vetting Q&A (Gemini deep-think, 2026-05-08, two passes)

Q1. Is the bound $\mathbb{P}(|F| > \lambda \|F\|_2) \leq 2 \exp(-c_d \lambda^{2/d})$ correct as stated? *Verdict: yes, exactly correct.* The prefactor 2 is the standard union bound for the two-sided absolute-value tail. The constant c_d is strictly universal and dimension-independent — this is the magic of the OU semigroup: hypercontractivity depends only on the chaos degree, not on the dimension of the Gaussian Hilbert space.

Q2. Cleanest reference? *Verdict: Janson, Theorem 5.10.* The gold standard, stated in exactly the abstract dimension-independent form needed. Adamczak-Wolff is correct but brings heavy machinery for non-Lipschitz functions that is not needed here. Stick to Janson.

Q3. Is the chain “polynomial chaos $\rightarrow$ tail bound $\rightarrow$ integrate to get $\int \exp(2|V|) \leq K$ ” the standard route? *Verdict: NO — invalid for $k \geq 2$.* The integrand $\exp(2\lambda - c\sqrt{\lambda})$ diverges. The polynomial-chaos bound is symmetric in $|F|$, but V_a is asymmetric (bounded below modulo Wick shifts, heavy right tail). What matters for $\int \exp(-2V_a)$ is the left tail of V_a alone, which the dynamical cutoff isolates by splitting $V = V_S + E_R$ and applying concentration only to E_R . See the Retraction in §4 above; Cluster A still needs the full Glimm-Jaffe Ch. 8 / Simon Ch. I structure.

Q4. Sharpness of $2/d$ for Wick polynomials. *Verdict: strictly saturated.* For large ϕ , the lower-order Wick subtractions are negligible, so $:\phi^d:$ $\sim \phi^d$ asymptotically. If $\phi \sim \mathcal{N}(0, 1)$, the tail of ϕ^d is exactly $\exp(-c\lambda^{2/d})$. There is no hidden stronger concentration that could rescue the divergent integral in Q3.

Q5. Existing Lean / Mathlib formalization? *Verdict: nothing in Mathlib.* `Mathlib.Probability.Distributions.Gaussian` has basic Gaussian measures, `Mathlib.Analysis.MeanInequalities` has L^p infrastructure, but the OU semigroup eigenfunction expansion and the Wiener chaos decomposition are entirely missing.

Q6. Coordination with **markov-semigroups**. *Verdict: yes — land the abstract theorem there.* `markov-semigroups` already has: - `MarkovSemigroups/Abstract/Hypercontractivity.lean`: `abstract IsHypercontractive +`

Gross’s equivalence between log-Sobolev and hypercontractivity (the LSI $\Rightarrow$ HC direction is currently axiomatized pending Bakry–Émery infrastructure). - `MarkovSemigroups/Abstract/LogSobolev.lean`: log-Sobolev inequality framework. - `MarkovSemigroups/Diffusion/OrnsteinUhlenbeck.lean`: abstract OU semigroup with Mehler’s formula, marked as Bakry–Émery with $\rho = \lambda_{\min}(C^{-1})$ — currently a 45-line skeleton. - `MarkovSemigroups/Diffusion/BakryEmery.lean`, `Diffusion/CarreDuChamp.lean`: Γ_2 calculus.

Missing in `markov-semigroups`: the Wiener chaos decomposition $L^2(\gamma_C) = \bigoplus_k \mathcal{H}_k$ with $\mathcal{H}_k$ as eigenspaces of the OU number operator (eigenvalue $-k$), and the explicit Hermite-polynomial / multilinear-functional models for $\mathcal{H}_k$. This is the missing-piece that needs to be built before the polynomial chaos concentration theorem can be stated, let alone proved.

Formalization strategy: finite-dim Hermite, not stochastic calculus

Per Gemini’s pass-2 advisory: although the abstract theorem (§2) is naturally stated for arbitrary Gaussian Hilbert spaces, the lattice approximation in `pphi2` keeps everything finite-dimensional. This buys a major strategic shortcut for the Lean formalization:

- Avoid Wiener–Itô / stochastic-calculus machinery. Multiple Wiener–Itô integrals require a substantial measure-theory and martingale prerequisite stack that Mathlib does not currently have. We do not need any of it.
- Define $\mathcal{H}_k$ algebraically via multivariate Hermite polynomials evaluated on the finite-dimensional Gaussian vector $(g_x)_{x \in \Lambda}$. Because the lattice GFF measure γ_C is non-degenerate on the finite lattice, the (probabilist’s) Hermite polynomials form a complete orthogonal basis of $L^2(\gamma_C)$, and the chaos decomposition $L^2(\gamma_C) = \bigoplus_k \mathcal{H}_k$ is the standard Hermite expansion.
- Prove the eigenfunction property of the finite-dim OU generator $L = \Delta - x \cdot \nabla$ directly on Hermite polynomials of total degree k (eigenvalue $-k$). This is a direct algebraic computation on multivariate polynomials and bypasses any infinite-dimensional spectral-theorem machinery.

The infinite-dimensional version of the abstract theorem is then a limit/closure step that we do not need to formalize to close the Cluster A axioms — every V_a on a fixed lattice lives in a finite chaos space.

A side benefit: the finite-dim Hermite + OU eigenfunction expansion is a clean self-contained Mathlib contribution, independent of the QFT application.

Time budget (revised after Gemini vetting)

The pre-vetting “3-day shortcut to discharge” was based on the integration chain Gemini retracted in Q3. Restoring the full dynamical-cutoff structure puts the realistic timeline back at the original Gemini estimate of 6-8 weeks, broken down as follows:

- Wiener chaos decomposition + OU eigenfunction expansion in `markov-semigroups`: $L^2(\gamma_C) = \bigoplus_k \mathcal{H}_k$ with $\mathcal{H}_k$ identified as eigenspaces of the OU number operator. Includes the multilinear-functional / Hermite-polynomial model for $\mathcal{H}_k$. ~1-2 weeks.
- Abstract polynomial chaos concentration theorem (Janson Thm 5.10) in `markov-semigroups`, derived from the existing abstract hypercontractivity
 - the new chaos decomposition. ~1-2 weeks.
- Specialize to the GFF in gaussian-field (or directly in `pphi2`): $V_a - \mathbb{E}V_a \in \mathcal{H}^{\leq 2k}$, $\|V_a\|_2 = O(1)$ uniformly in N via the existing `CovarianceSplit.lean` + `HeatKernelBound.lean` infrastructure. ~3-5 days.
- Wire the dynamical cutoff in `pphi2/NelsonEstimate/`: replace the `True`-stub bodies of `rough_error_Lp_bound` / `rough_error_tail_bound`; combine with the already-proved `smooth_interaction_lower_bound_log` and the new chaos concentration theorem; do the $T = T(M)$ optimization in `NelsonEstimate.lean`. ~2-3 weeks — this is the bottleneck: Gemini flags heavy measure-theoretic friction in the domain splitting $\{|\phi_R| \leq R\}$ vs $\{|\phi_R| > R\}$, Gaussian tail bounds on the latter, and recombination of the indicator-weighted integrals. Worth investing in reusable Mathlib-style domain-splitting + Gaussian-tail-integration

lemmas, since the same friction recurs in any future CQFT-style estimate (Phase 4 `pphi2_nontriviality`, $O(N)$ sigma-model mass gap, etc.).

- Specialize to the four Cluster A axioms (lattice / asym lattice flavours, with and without the hypercontractive normalization): ~2-3 days.

Total: 6-8 weeks, matching the original Gemini estimate.

Two reasons this investment pays back beyond Cluster A:

1. **markov-semigroups** becomes Mathlib-ready in the OU/hypercontractivity branch. Janson Theorem 5.10 in Lean 4 is a substantial probability-library contribution in its own right — there is currently nothing comparable upstream — and the Wiener-chaos / Hermite-eigenfunction formalization is reusable wherever Gaussian polynomial concentration matters.
2. The dynamical-cutoff machinery transfers to other CQFT estimates: the same domain-splitting + Gaussian-tail-integration scaffolding shows up in Phase 4 (`pphi2_nontriviality`, the genuinely non-Gaussian discharge) and in any future $O(N)$ sigma-model / Yukawa work in `pphi2N` or sister projects.